

Optimization 2

ISE 5406

Lecture 11b: Convergence of Gradient Descent on Smooth, Strongly Convex Functions

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Where We Are in the Course

Last Time (Lecture 11): Steepest Descent & Newton's Methods

- Steepest descent for multivariate problems
- Inexact line search (Wolfe conditions)
- Newton's method for multivariate optimization
- Modifications: Levenberg–Marquardt, quasi-Newton methods

Today (Lecture 11b): Convergence Analysis for Smooth, Strongly Convex Functions

- Recap: L -smooth functions and the descent lemma
- Recap: μ -strongly convex functions
- Convergence of GD on convex, L -smooth functions ($O(1/T)$ rate)
- Convergence of GD on μ -strongly convex, L -smooth functions (linear rate)
- Condition number and its role

Next Time (Lecture 12): Conjugate Direction Methods

- Conjugate directions: a powerful concept for quadratics
- Conjugate gradient method
- Convergence in at most n steps

Big Picture: Today we give a rigorous convergence analysis for gradient descent under two key structural assumptions — smoothness and strong convexity. These results explain **why** gradient descent works and **how fast** it converges.

L -Smooth Functions (Recap)

Convergence of GD: Convex + L -Smooth

μ -Strongly Convex Functions (Recap)

Convergence of GD: L -Smooth + μ -Strongly Convex

Condition Number and Convergence

Gradient Flow Perspective (Optional)

Summary

L-Smooth Functions (Recap)

Convergence of GD: Convex + *L*-Smooth

μ -Strongly Convex Functions (Recap)

Convergence of GD: *L*-Smooth + μ -Strongly Convex

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Summary

Lipschitz Continuity

Definition (Lipschitz Continuity)

A function $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **L -Lipschitz** if for all

$\mathbf{x}, \mathbf{y} \in \text{dom } g$,

$$\|g(\mathbf{x}) - g(\mathbf{y})\| \leq L\|\mathbf{x} - \mathbf{y}\|.$$

Examples:

- $f(\mathbf{x}) = \mathbf{w}^T \mathbf{x}$ is $\|\mathbf{w}\|$ -Lipschitz.
- $f(\mathbf{x}) = Q\mathbf{x}$ where Q is symmetric is $\max\{|\lambda_{\min}(Q)|, |\lambda_{\max}(Q)|\}$ -Lipschitz.

Intuition: A Lipschitz function cannot change faster than a fixed linear rate. There exists a “double cone” that can slide along the graph with the whole function staying outside it.

L -Smooth Functions: Definition

Definition (Smoothness)

A differentiable function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is L -**smooth** if ∇f is L -Lipschitz, i.e., for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$,

$$\|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\| \leq L\|\mathbf{x} - \mathbf{y}\|.$$

Example: $f(\mathbf{x}) = \mathbf{x}^T Q \mathbf{x}$ with $Q \succeq 0$ is $2\lambda_{\max}(Q)$ -smooth, since $\nabla f(\mathbf{x}) = 2Q\mathbf{x}$.

Intuition: The gradient does not change too rapidly. This rules out functions like $f(x) = |x|$ where ∇f has a discontinuity at the origin.

Equivalent Characterizations of Smoothness

Lemma (Hessian Characterization)

Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is twice differentiable. Then f is L -smooth if and only if

$$-LI_n \preceq \nabla^2 f(\mathbf{x}) \preceq LI_n \quad \text{for all } \mathbf{x} \in \mathbb{R}^n.$$

Equivalently, $|\lambda_i(\nabla^2 f(\mathbf{x}))| \leq L$ for all eigenvalues and all $\mathbf{x}$.

Lemma (Function Value Characterization)

f is L -smooth if and only if for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$,

$$|f(\mathbf{y}) - f(\mathbf{x}) - \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x})| \leq \frac{L}{2} \|\mathbf{y} - \mathbf{x}\|^2.$$

Key takeaway: For L -smooth functions, f is sandwiched between two quadratics centered at any point — one from convexity (below), one from smoothness (above).

The Descent Lemma

Theorem (Descent Lemma)

For an L -smooth differentiable function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ (not necessarily convex), if the gradient descent step size satisfies $\eta \leq 1/L$, then

$$f(\mathbf{x}_{k+1}) \leq f(\mathbf{x}_k) - \frac{\eta}{2} \|\nabla f(\mathbf{x}_k)\|^2.$$

Proof sketch: By L -smoothness applied to $\mathbf{y} = \mathbf{x}_k - \eta \nabla f(\mathbf{x}_k)$:

$$\begin{aligned} f(\mathbf{x}_{k+1}) &\leq f(\mathbf{x}_k) - \eta \|\nabla f(\mathbf{x}_k)\|^2 + \frac{L\eta^2}{2} \|\nabla f(\mathbf{x}_k)\|^2 \\ &= f(\mathbf{x}_k) - \eta \left(1 - \frac{L\eta}{2}\right) \|\nabla f(\mathbf{x}_k)\|^2 \\ &\leq f(\mathbf{x}_k) - \frac{\eta}{2} \|\nabla f(\mathbf{x}_k)\|^2 \end{aligned}$$

where the last step uses $\eta \leq 1/L$, so $1 - L\eta/2 \geq 1/2$.

Each step makes guaranteed progress proportional to $\|\nabla f(\mathbf{x}_k)\|^2$.

L -Smooth Functions (Recap)

Convergence of GD: Convex + L -Smooth

μ -Strongly Convex Functions (Recap)

Convergence of GD: L -Smooth + μ -Strongly Convex

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Summary

Gradient Descent Algorithm (Fixed Step Size)

Algorithm: Gradient Descent with Fixed Step Size

Given: $f : \mathbb{R}^n \rightarrow \mathbb{R}$ convex and L -smooth, step size $\eta \leq 1/L$, starting point $\mathbf{x}_0$.

For $k = 0, 1, 2, \dots, T - 1$:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \eta \nabla f(\mathbf{x}_k)$$

Key questions:

- Does $f(\mathbf{x}_k) \rightarrow f^*$? **Yes** (by the descent lemma + convexity)
- How fast? **This is the convergence rate question**
- How does the rate depend on the problem structure?

$O(1/T)$ Convergence Rate

Theorem (Convergence: Convex + L -Smooth)

Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex and L -smooth. Choose

$\eta \leq 1/L$. Then after T steps of gradient descent:

$$f(\mathbf{x}_T) - f(\mathbf{x}^*) \leq \frac{\|\mathbf{x}_0 - \mathbf{x}^*\|^2}{2T\eta}.$$

In particular, with $\eta = 1/L$: $f(\mathbf{x}_T) - f^* \leq \frac{L\|\mathbf{x}_0 - \mathbf{x}^*\|^2}{2T}.$

Interpretation:

- To achieve $f(\mathbf{x}_T) - f^* \leq \varepsilon$, we need $T = O(1/\varepsilon)$ iterations.
- This is a **sublinear** rate — progress slows down over time.
- The rate depends on the initial distance $\|\mathbf{x}_0 - \mathbf{x}^*\|$.

Proof: $O(1/T)$ Rate

Step 1: Analyze $\|\mathbf{x}_{k+1} - \mathbf{x}^*\|^2$:

$$\begin{aligned}\|\mathbf{x}_{k+1} - \mathbf{x}^*\|^2 &= \|\mathbf{x}_k - \eta \nabla f(\mathbf{x}_k) - \mathbf{x}^*\|^2 \\ &= \|\mathbf{x}_k - \mathbf{x}^*\|^2 - 2\eta \nabla f(\mathbf{x}_k)^T (\mathbf{x}_k - \mathbf{x}^*) + \eta^2 \|\nabla f(\mathbf{x}_k)\|^2.\end{aligned}$$

Step 2: Bound each term:

- By the **descent lemma**: $\eta^2 \|\nabla f(\mathbf{x}_k)\|^2 \leq 2\eta(f(\mathbf{x}_k) - f(\mathbf{x}_{k+1}))$
(since $\eta \leq 1/L$).
- By **convexity**: $-2\eta \nabla f(\mathbf{x}_k)^T (\mathbf{x}_k - \mathbf{x}^*) \leq 2\eta(f(\mathbf{x}^*) - f(\mathbf{x}_k))$.

Step 3: Combine:

$$\|\mathbf{x}_{k+1} - \mathbf{x}^*\|^2 \leq \|\mathbf{x}_k - \mathbf{x}^*\|^2 + 2\eta(f(\mathbf{x}^*) - f(\mathbf{x}_{k+1})).$$

Step 4: Telescope from $k = 0$ to $T - 1$:

$$\|\mathbf{x}_T - \mathbf{x}^*\|^2 - \|\mathbf{x}_0 - \mathbf{x}^*\|^2 \leq 2\eta \sum_{k=0}^{T-1} (f(\mathbf{x}^*) - f(\mathbf{x}_k)) \leq 2T\eta(f(\mathbf{x}^*) - f(\mathbf{x}_T)).$$

Since $\|\mathbf{x}_T - \mathbf{x}^*\|^2 \geq 0$, we conclude $f(\mathbf{x}_T) - f^* \leq \frac{\|\mathbf{x}_0 - \mathbf{x}^*\|^2}{2T\eta}$. $\square$

Can We Do Better?

The $O(1/T)$ rate is disappointing:

- To double accuracy, we must double the number of iterations.
- For $\varepsilon = 10^{-6}$, we need $T \sim 10^6$ iterations!

What went wrong? Consider $f(x) = -\log x$ (truncated). It is convex and 1-smooth (for $x \geq 1$), but:

$$x_{k+1} = x_k + \eta/x_k \leq x_k + 1/x_k$$

so convergence is very slow when $x_0 = 1$.

Key Question

Under which additional assumptions can gradient descent converge **exponentially fast** (i.e., a **linear convergence rate**)?

Answer: **Strong convexity** — the function curves at least as much as some quadratic.

L -Smooth Functions (Recap)

Convergence of GD: Convex + L -Smooth

μ -Strongly Convex Functions (Recap)

Convergence of GD: L -Smooth + μ -Strongly Convex

Condition Number and Convergence

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Summary

Strong Convexity: Definition

Definition (Strong Convexity)

A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is μ -strongly convex (with $\mu > 0$)

if

$$f(\mathbf{x}) - \frac{\mu}{2} \|\mathbf{x}\|^2 \text{ is convex.}$$

Equivalent characterizations:

- **Second-order:** If f is twice differentiable, then f is μ -strongly convex iff $\nabla^2 f(\mathbf{x}) \succeq \mu I_n$ for all $\mathbf{x}$.
- **First-order:** f is μ -strongly convex iff for all $\mathbf{x}, \mathbf{y}$,

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T (\mathbf{y} - \mathbf{x}) + \frac{\mu}{2} \|\mathbf{y} - \mathbf{x}\|^2.$$

- **Monotone gradients:** f is μ -strongly convex iff

$$(\nabla f(\mathbf{x}) - \nabla f(\mathbf{y}))^T (\mathbf{x} - \mathbf{y}) \geq \mu \|\mathbf{x} - \mathbf{y}\|^2.$$

Strong Convexity: Examples

Examples

1. $f(\mathbf{x}) = a\|\mathbf{x}\|^2$, $a > 0 \Rightarrow 2a$ -strongly convex.
2. $f(\mathbf{x}) = \mathbf{x}^T Q \mathbf{x}$, $Q \succ 0 \Rightarrow 2\lambda_{\min}(Q)$ -strongly convex.
3. $f(x) = -\log x$ is **not** strongly convex ($f''(x) = 1/x^2 \rightarrow 0$).
4. $f(x) = x^4$ is **not** strongly convex ($f''(0) = 0$).
5. Affine functions $f(\mathbf{x}) = \mathbf{a}^T \mathbf{x} + b$ are **not** strongly convex.

Geometric interpretation: A μ -strongly convex function is bounded below by a quadratic “bowl” at every point:

$$f(\mathbf{y}) \geq \underbrace{f(\mathbf{x}) + \nabla f(\mathbf{x})^T (\mathbf{y} - \mathbf{x})}_{\text{tangent hyperplane}} + \underbrace{\frac{\mu}{2} \|\mathbf{y} - \mathbf{x}\|^2}_{\text{quadratic curvature}}.$$

This means the function **curves away from its tangent planes** at a uniform rate.

The Polyak–Łojasiewicz (PL) Inequality

Theorem (Polyak–Łojasiewicz Inequality)

If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is μ -strongly convex, then for all $\mathbf{x} \in \mathbb{R}^n$:

$$\|\nabla f(\mathbf{x})\|^2 \geq 2\mu(f(\mathbf{x}) - f(\mathbf{x}^*)).$$

Proof: By μ -strong convexity with $\mathbf{y} = \mathbf{x}^*$ and using $\nabla f(\mathbf{x}^*) = \mathbf{0}$:

$$f(\mathbf{x}) - f(\mathbf{x}^*) \geq \frac{\mu}{2} \|\mathbf{x} - \mathbf{x}^*\|^2.$$

By convexity:

$$f(\mathbf{x}) - f(\mathbf{x}^*) \leq \nabla f(\mathbf{x})^T (\mathbf{x} - \mathbf{x}^*) \leq \|\nabla f(\mathbf{x})\| \cdot \|\mathbf{x} - \mathbf{x}^*\|.$$

Combining:
$$f(\mathbf{x}) - f(\mathbf{x}^*) \leq \|\nabla f(\mathbf{x})\| \cdot \sqrt{\frac{2}{\mu}(f(\mathbf{x}) - f(\mathbf{x}^*))}.$$

Squaring both sides and dividing by $f(\mathbf{x}) - f(\mathbf{x}^*)$ yields:

$$f(\mathbf{x}) - f(\mathbf{x}^*) \leq \frac{1}{2\mu} \|\nabla f(\mathbf{x})\|^2. \quad \square$$

The PL inequality says: large gradient $\Rightarrow$ large suboptimality gap. This is the key ingredient for proving linear convergence!

L -Smooth Functions (Recap)

Convergence of GD: Convex + L -Smooth

μ -Strongly Convex Functions (Recap)

Convergence of GD: L -Smooth + μ -Strongly Convex

Condition Number and Convergence

Gradient Flow Perspective (Optional)

Summary

The Main Convergence Theorem

Theorem (Linear Convergence Rate)

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be μ -strongly convex and L -smooth.

Consider gradient descent with step size $\lambda = 1/L$:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \frac{1}{L} \nabla f(\mathbf{x}_k).$$

Then:

$$f(\mathbf{x}_k) - f(\mathbf{x}^*) \leq \left(1 - \frac{\mu}{L}\right)^{k-1} (f(\mathbf{x}_1) - f(\mathbf{x}^*)) = \left(1 - \frac{1}{\kappa}\right)^{k-1} (f(\mathbf{x}_1) - f(\mathbf{x}^*)).$$

This is exponential (linear) convergence!

- Each iteration reduces the suboptimality gap by a factor of $1 - 1/\kappa$.
- To achieve $f(\mathbf{x}_T) - f^* \leq \varepsilon$: need $T = O(\kappa \log \frac{1}{\varepsilon})$ iterations.
- Compare with $O(1/\varepsilon)$ for the merely convex case!

Proof: Linear Convergence

The proof combines just two ingredients: the [descent lemma](#) and the [PL inequality](#).

Step 1: Descent lemma (proved earlier): with $\lambda = 1/L$,

$$f(\mathbf{x}_{k+1}) \leq f(\mathbf{x}_k) - \frac{1}{2L} \|\nabla f(\mathbf{x}_k)\|^2.$$

Step 2: Apply the PL inequality:

$$\|\nabla f(\mathbf{x}_k)\|^2 \geq 2\mu(f(\mathbf{x}_k) - f(\mathbf{x}^*)).$$

Substituting:

$$f(\mathbf{x}_{k+1}) \leq f(\mathbf{x}_k) - \frac{1}{2L} \cdot 2\mu(f(\mathbf{x}_k) - f(\mathbf{x}^*)) = f(\mathbf{x}_k) - \frac{\mu}{L}(f(\mathbf{x}_k) - f(\mathbf{x}^*)).$$

Step 3: Subtract $f(\mathbf{x}^*)$ from both sides:

$$f(\mathbf{x}_{k+1}) - f(\mathbf{x}^*) \leq \left(1 - \frac{\mu}{L}\right) (f(\mathbf{x}_k) - f(\mathbf{x}^*)) = \left(1 - \frac{1}{\kappa}\right) (f(\mathbf{x}_k) - f(\mathbf{x}^*)).$$

Step 4: Apply recursively:

$$f(\mathbf{x}_k) - f(\mathbf{x}^*) \leq \left(1 - \frac{1}{\kappa}\right)^{k-1} (f(\mathbf{x}_1) - f(\mathbf{x}^*)). \quad \square$$

The entire proof is: **descent lemma + PL $\Rightarrow$ contraction $\Rightarrow$ recurse.**

Iterate Convergence (Corollary)

Corollary (Iterate Convergence)

Under the same assumptions:

$$\|\mathbf{x}_k - \mathbf{x}^*\|^2 \leq \left(1 - \frac{1}{\kappa}\right)^{k-1} \|\mathbf{x}_1 - \mathbf{x}^*\|^2.$$

Proof: By μ -strong convexity (using $\nabla f(\mathbf{x}^*) = \mathbf{0}$):

$$f(\mathbf{x}_k) - f(\mathbf{x}^*) \geq \frac{\mu}{2} \|\mathbf{x}_k - \mathbf{x}^*\|^2.$$

Combining with the function value bound:

$$\frac{\mu}{2} \|\mathbf{x}_k - \mathbf{x}^*\|^2 \leq f(\mathbf{x}_k) - f(\mathbf{x}^*) \leq \left(1 - \frac{1}{\kappa}\right)^{k-1} (f(\mathbf{x}_1) - f(\mathbf{x}^*)).$$

Since $f(\mathbf{x}_1) - f(\mathbf{x}^*) \leq \frac{L}{2} \|\mathbf{x}_1 - \mathbf{x}^*\|^2$ (by L -smoothness with $\nabla f(\mathbf{x}^*) = \mathbf{0}$):

$$\|\mathbf{x}_k - \mathbf{x}^*\|^2 \leq \frac{L}{\mu} \left(1 - \frac{1}{\kappa}\right)^{k-1} \|\mathbf{x}_1 - \mathbf{x}^*\|^2 = \kappa \left(1 - \frac{1}{\kappa}\right)^{k-1} \|\mathbf{x}_1 - \mathbf{x}^*\|^2.$$

Remark: The leading constant κ can be absorbed; the dominant behavior is still the geometric decay $(1 - 1/\kappa)^k$.

Comparison with Quadratic Analysis

Connecting to Earlier Results

For the quadratic $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T Q \mathbf{x} + \mathbf{c}^T \mathbf{x}$ with $Q \succ 0$:

- $L = \lambda_{\max}(Q)$ and $\mu = \lambda_{\min}(Q)$
- Condition number: $\kappa = \lambda_{\max}(Q)/\lambda_{\min}(Q)$
- Convergence factor: $1 - 1/\kappa = 1 - \lambda_{\min}(Q)/\lambda_{\max}(Q)$

This matches our earlier quadratic convergence bound exactly!

Key insight: The PL-based proof reveals **why** the same rate applies:

- The descent lemma captures the “upper curvature” (L -smoothness).
- The PL inequality captures the “lower curvature” (μ -strong convexity).
- Together, they sandwich the progress per step between two quadratics.

Remark: The PL inequality actually holds for a broader class of functions than just strongly convex ones. Any function satisfying PL enjoys linear convergence under GD — this is an active area of research!

L -Smooth Functions (Recap)

Convergence of GD: Convex + L -Smooth

μ -Strongly Convex Functions (Recap)

Convergence of GD: L -Smooth + μ -Strongly Convex

Condition Number and Convergence

Gradient Flow Perspective (Optional)

Summary

The Condition Number

Definition (Condition Number)

For an L -smooth, μ -strongly convex function f , the **condition number** is:

$$\kappa = \frac{L}{\mu} \geq 1.$$

For a quadratic $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T Q \mathbf{x}$ with $Q \succ 0$: $\kappa = \frac{\lambda_{\max}(Q)}{\lambda_{\min}(Q)}$.

Convergence rate in terms of κ :

$$\|\mathbf{x}_T - \mathbf{x}^*\|^2 \leq \left(1 - \frac{1}{\kappa}\right)^T \|\mathbf{x}_0 - \mathbf{x}^*\|^2.$$

Iteration complexity: To achieve $\|\mathbf{x}_T - \mathbf{x}^*\|^2 \leq \varepsilon \|\mathbf{x}_0 - \mathbf{x}^*\|^2$:

$$T \geq \kappa \ln(1/\varepsilon).$$

A large condition number κ means slow convergence!

Condition Number: Examples on Quadratics

Recall: For $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T Q \mathbf{x}$ with $Q \succ 0$, the optimal step size is

$$\eta = \frac{2}{\lambda_{\min} + \lambda_{\max}}, \text{ giving:}$$

$$\|\mathbf{x}_k - \mathbf{x}^*\|^2 \leq \left(\frac{\lambda_{\max} - \lambda_{\min}}{\lambda_{\max} + \lambda_{\min}} \right)^{2k} \|\mathbf{x}_0 - \mathbf{x}^*\|^2 = \left(\frac{\kappa - 1}{\kappa + 1} \right)^{2k} \|\mathbf{x}_0 - \mathbf{x}^*\|^2.$$

Examples

- $Q = \text{diag}(1/2, 1)$: $\kappa = 2$, convergence rate = $1/3$. **Fast!**
- $Q = \text{diag}(1/100, 1)$: $\kappa = 100$, convergence rate = $99/101 \approx 0.98$. **Slow!**

Geometric picture:

- Well-conditioned (κ small): nearly circular level sets $\Rightarrow$ fast convergence.
- Ill-conditioned (κ large): elongated elliptical level sets $\Rightarrow$ zig-zagging.

Why Condition Number Matters: Non-Quadratic Functions

For a general (non-quadratic) L -smooth, μ -strongly convex function, near the minimizer $\mathbf{x}^*$ we can approximate:

$$f(\mathbf{x}) \approx f(\mathbf{x}^*) + \frac{1}{2}(\mathbf{x} - \mathbf{x}^*)^T \nabla^2 f(\mathbf{x}^*)(\mathbf{x} - \mathbf{x}^*).$$

So the **local** convergence rate depends on:

$$\kappa(\nabla^2 f(\mathbf{x}^*)) = \frac{\lambda_{\max}(\nabla^2 f(\mathbf{x}^*))}{\lambda_{\min}(\nabla^2 f(\mathbf{x}^*))}.$$

Summary of convergence rates:

Assumption	Rate	Iterations for ε -accuracy
Convex + L -smooth	$O(1/T)$	$O(1/\varepsilon)$
+ μ -strongly convex	$O((1 - \mu/L)^T)$	$O(\kappa \log(1/\varepsilon))$

Strong convexity upgrades the rate from sublinear to linear (exponential).

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Convergence of GD: L -Smooth + μ -Strongly Convex

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Gradient Flow Perspective (Optional)

Summary

Gradient Flow: Continuous-Time Gradient Descent

Definition (Gradient Flow)

Given a smooth convex f and initial point $\hat{\mathbf{x}}$, the **gradient flow** is the solution to:

$$\frac{d}{dt}\mathbf{x}_t = -\nabla f(\mathbf{x}_t), \quad \mathbf{x}_0 = \hat{\mathbf{x}}.$$

Key property: $f(\mathbf{x}_t)$ is decreasing, since:

$$\frac{d}{dt}f(\mathbf{x}_t) = \nabla f(\mathbf{x}_t)^T \dot{\mathbf{x}}_t = -\|\nabla f(\mathbf{x}_t)\|^2 \leq 0.$$

With strong convexity: We can show exponential decay:

$$\frac{d}{dt}\|\mathbf{x}_t - \mathbf{x}^*\|^2 = -2\nabla f(\mathbf{x}_t)^T(\mathbf{x}_t - \mathbf{x}^*) \leq -2\mu\|\mathbf{x}_t - \mathbf{x}^*\|^2$$

by the monotone gradient characterization. By Gronwall's lemma:

$$\|\mathbf{x}_T - \mathbf{x}^*\|^2 \leq e^{-2\mu T}\|\mathbf{x}_0 - \mathbf{x}^*\|^2.$$

Discrete GD ($1 - \mu\eta$ per step) $\approx$ discretization of continuous flow ($e^{-2\mu T}$).

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μ -Strongly Convex Functions (Recap)

Convergence of GD: L -Smooth + μ -Strongly Convex

Condition Number and Convergence

Gradient Flow Perspective (Optional)

Summary

Summary: Key Results

What We Proved Today

1. **Descent Lemma:** L -smoothness guarantees
$$f(\mathbf{x}_{k+1}) \leq f(\mathbf{x}_k) - \frac{\eta}{2} \|\nabla f(\mathbf{x}_k)\|^2.$$
2. **Convex + L -smooth:** GD converges at rate $O(1/T)$.
3. **μ -strongly convex + L -smooth:** GD converges at **linear rate** $(1 - \mu/L)^T$.
4. **Condition number** $\kappa = L/\mu$ controls the convergence speed.

Key Takeaways

- Smoothness lets us choose a constant step size and get guaranteed descent.
- Strong convexity provides a “lower curvature bound” that prevents slow convergence.
- The ratio $\kappa = L/\mu$ is the fundamental quantity governing GD performance.
- Ill-conditioned problems ($\kappa \gg 1$) require more sophisticated methods (e.g., Newton, conjugate gradient, preconditioning).

Next Lecture (Lecture 12): Conjugate Direction Methods

- **Conjugate directions**: A powerful concept for quadratics
- **Conjugate gradient method**: Combines efficiency of Newton with simplicity of SD
- Convergence in at most n steps (no rounding!)
- Extends to nonlinear problems

Why care?

- Conjugate gradient achieves **better dependence on κ** : rate $\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^T$ vs. $\left(\frac{\kappa-1}{\kappa+1}\right)^T$ for GD.
- Bridges gap between simple (Steepest Descent) and powerful (Newton) methods
- Foundation for modern optimization solvers

Questions?

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